

ticularly malicious, for once installed they'll open up avenues of entries for other attackers. Thus, once a system is identified to be compromised by a rootkit, it is normal to reinstall it from scratch. Emphasis is laid on reinstallation because post a rootkit attack, even the most experienced and weathered of systems or security engineers might not be able to eradicate its repercussions.

- **Binary Verification:** Carrying out the verification of binaries and important files is considered an essential step in the security process. It keeps the security teams and the system administration posted regarding all updates concerning system changes. The software application entrusted with monitoring the system for said changes is known as an Intrusion Detection System (IDS). One of the most redeeming aspects of FreeBSD is that it offers native support for a fundamental IDS system.
- **System Tuning For Security:** Through the use of `sysctl`, FreeBSD allows for the tuning of many of its system features. With the use of `sysctl`, security features responsible for blocking Denial of Service (DoS) attacks can also be tuned.

One-time Passwords

As such, FreeBSD contains support for One-time Passwords In Everything (OPIE). The objective of OPIE is to keep replay attacks at bay, which would otherwise enable an attacker to discover the password of a user and subsequently make use of it to break into a system. However, under OPIE, since a password can be utilised only once, a discovered password no longer presents any threat to the system. To manage passwords, OPIE uses a challenge/response system in conjunction with a secure hash. There are three different types of passwords that OPIE uses.

The first kind is the usual Kerberos password. The second is the one-time password that is produced by 'opiekey'. Lastly, the third type is the 'secret password' which is utilised to create one-time passwords. The two bits of crucial data for OPIE are the 'seed' or the 'key', and the 'iteration count'. To generate the one-time password, OPIE uses a complicated algorithm that

```
$ opiepasswd -c
Adding entry:
Only use this method from the console: NEVER from remote. If you are using
telnet, xterm, or a dial-in, type ^C now or exit with no password.
Then run opiepasswd without the -c parameter.
Using MD5 to compute responses.
Enter new secret pass phrase:
Again new secret pass phrase:
```

A look at the OPIE console

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